

3. Data Cleaning

3.1 Data Source

This study uses data from **Afrobarometer Round 9**, a nationally representative survey conducted across African countries to measure citizens' experiences, perceptions, and access to public services. The analysis focuses on a **SADC-focused sample of 16 countries**, including Angola, Botswana, Eswatini, Kenya, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, Seychelles, South Africa, Tanzania, Uganda, Zambia, and Zimbabwe. The final dataset contains **23,076 respondents**, providing broad regional coverage across Southern and Africa.

3.2 Variables Used in the Analysis

The study combines two distinct types of information available in Afrobarometer.

First, interviewer observations were used to identify whether public services were physically present within the respondent's community. These variables included the presence of:

- Piped water systems (EA_SVC_B)
- Health clinics (EA_FAC_D)
- Electricity grids (EA_SVC_A)
- Financial services such as banks, mobile money services, or ATMs (EA_FAC_F)

Second, respondents reported the frequency with which they experienced shortages of essential services during the previous year. These variables included:

- Water shortages (Q6B)
- Medical care shortages (Q6C)
- Fuel shortages (Q6D)
- Cash-income shortages (Q6E)

By combining interviewer observations with respondent experiences, the study evaluates whether the visible presence of infrastructure corresponds to reliable service delivery.

3.3 Data Cleaning

Before analysis, individual Afrobarometer Round 9 country files were merged to create one working dataset for the SADC-focused sample. The merge was completed in three stages. First, country files from Angola to Mauritius were read, selected, cleaned, and combined. Second, country files from Mozambique to Zimbabwe were processed in the same way. Finally, both merged files were combined into a single working dataset.

MERGED FILES link:

https://drive.google.com/drive/folders/14URPH5_fuiFkOQXwseAagfHN4cT_Flaq?usp=drive_link

Afrobarometer variables contain value labels and non-response categories that must be removed before statistical analysis. All service and shortage variables were converted to numeric format and restricted to valid response ranges. Responses outside the expected ranges were treated as missing values.

A custom cleaning function was applied to standardize variables and remove invalid observations. This ensured that all mismatch calculations were based only on valid survey responses.

R Code (please refer to the uploaded Data preparation and merging R script)

This code merged the 16 Afrobarometer country datasets, retained only the variables required for the analysis, removed SPSS value labels using `zap_labels()`, and added country identifiers through a file-name lookup table before creating the final analysis dataset.

3.4 Construction of Severe Shortage Variables

The original Afrobarometer shortage questions use a five-point scale ranging from 0 (Never) to 4 (Always). To identify severe deprivation, binary indicators were created for each service category. Respondents reporting shortages several times, many times, or always (values 2–4) were classified as experiencing a severe shortage, while those reporting shortages never or only once or twice (values 0–1) were classified as not experiencing a severe shortage. This transformation allows the analysis to focus on meaningful service failures rather than occasional disruptions.

R Code (please refer to the uploaded Data cleaning and variable construction R script)

3.5 Construction of Public Service Visibility Mismatch Variables

The central concept of the study is the **Public Service Visibility Mismatch**.

A mismatch occurs when a public service is **visibly present** within a community, but respondents continue to report a **severe shortage** of that service.

For example:

- A water mismatch occurs when piped water infrastructure is observed but severe water shortages are still reported.
- A health mismatch occurs when a health clinic is present but respondents report severe difficulties obtaining medical care.
- An electricity-fuel mismatch occurs when an electricity grid exists but severe fuel or energy shortages remain common.
- A cash-access mismatch occurs when banking or mobile money services are available but respondents continue to experience severe cash shortages.

These mismatch indicators form the foundation of all descriptive, comparative, and regression analyses presented in the study.

R Code (please refer to the uploaded Data cleaning and variable construction R script)

For the mismatch variables, the stricter definition was used: a mismatch occurs when the service is present but the respondent reports the shortage as “always” occurring.